
Dependency-Aware Multi-Phase Liver Tumor Segmentation with Explicit Phase Interaction Modeling

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GitHub Repository Link: <https://github.com/areebanaveed-12/Phase-Dependency-Liver-Segmentation>

Abstract

Liver tumor segmentation from CT is clinically important for diagnosis, treatment planning, and longitudinal monitoring. We present a dependency-aware framework for multi-phase CT segmentation that addresses the gap, identified in prior state-of-the-art surveys, between implicit fusion methods and explicit modeling of phase relationships. Building on a single-phase U-Net baseline on LiTS and a four-phase attention U-Net baseline on Harvard multi-phase CT, we introduce a segmentation model that preserves phase-specific representations, learns per-phase importance gates, and incorporates an explicit pairwise cross-phase interaction term prior to decoding. The proposed system operates on a 2.5D input with three neighboring slices per phase, employs tumor-aware sampling, a combined focal cross-entropy, Dice, and focal Tversky objective, and validation-tuned inference thresholds. On a 100-patient Harvard cohort, the model attains a liver Dice score of 0.9435 and a tumor Dice score of 0.6975. A central contribution is a complete combinational dependency analysis on the fixed trained model across all 15 possible non-empty phase combinations. The analysis reveals that the best two-phase subset, PVP+DP, achieves a tumor Dice of 0.7209 and the best three-phase subset, AP+PVP+DP, achieves 0.7109, both surpassing the all-phase configuration at 0.6975.

1. Introduction

Liver tumor segmentation from CT lies at the intersection of medical image analysis and clinical decision support. Accurate delineation of the liver and intrahepatic tumors matters because it affects diagnosis, treatment planning, volumetric assessment, and longitudinal monitoring. In routine abdominal imaging, the problem is not limited to a single scan. Multi-phase CT acquires the liver across non-contrast (NC), arterial phase (AP), portal venous phase (PVP), and delayed

phase (DP), and each phase reveals different enhancement behavior and lesion conspicuity (Wu et al., 2024; Gul et al., 2022; Chernyak et al., 2018).

A prior state-of-the-art survey conducted as part of this work helped to establish the problem domain, characterise the principal segmentation baselines, and isolate the remaining open challenge. Classical encoder-decoder architectures such as U-Net are effective for dense medical segmentation (Ronneberger et al., 2015), and more recent multi-phase methods improve results through attention, fusion, adaptive weighting, or missing-phase compensation (Wu et al., 2024; Gul et al., 2022; Zhang et al., 2021b; Xu et al., 2021; Liu et al., 2024; Hu et al., 2024). However, many of these methods continue to treat phase relationships implicitly: they exploit multiple phases without directly examining which combinations are most informative or how phases interact.

This observation motivates the central question of the present work. Rather than asking only whether multi-phase input improves segmentation, we ask whether a model can preserve phase-specific information long enough to reason about cross-phase dependence explicitly. The work is therefore organised as a methodological progression: Phase 3 establishes a corrected single-phase LiTS baseline, Phase 4 adapts the pipeline to Harvard multi-phase CT, and Phase 5 introduces a dependency-aware model together with a complete combinational dependency analysis on the fixed trained model.

Contributions:

1. Establish a reliable segmentation baseline using a corrected single-phase U-Net on LiTS, providing a concrete reference point against which later improvements can be interpreted.
2. Adapt the pipeline from single-phase CT to a realistic four-phase Harvard setting with phase-aware pre-processing, caching, and patient-wise evaluation.
3. Introduce a dependency-aware multi-phase architecture that preserves per-phase representations, learns

phase-specific importance, and models explicit cross-phase interactions.

4. Evaluate not only final Dice scores but also phase behaviour through a complete study of all 15 possible non-empty phase combinations.
5. Demonstrate that certain reduced phase subsets can outperform the all-phase configuration, indicating that explicit dependency analysis is a substantive contribution rather than a cosmetic addition.

2. Related Work

U-Net remains the canonical encoder–decoder baseline for dense medical image segmentation (Ronneberger et al., 2015), and LiTS remains the most widely recognized public benchmark for liver and liver tumor segmentation (Bilic et al., 2023). Task-specific liver segmentation work such as H-DenseUNet (Li et al., 2018), together with modern segmentation systems such as nnU-Net (Isensee et al., 2021), UNETR (Hatamizadeh et al., 2022), and Swin-UNETR (Tang et al., 2022), established the importance of strong feature hierarchies, context modeling, and robust training pipelines.

Multi-phase methods have demonstrated that richer CT input can improve segmentation through channel fusion, phase weighting, spatial aggregation, or image synthesis (Wu et al., 2024; Gul et al., 2022; Zhang et al., 2021b; Xu et al., 2021; Raju et al., 2020; Liu et al., 2024). The principal open issue, however, is no longer whether multiple phases help, but how their relationships should be modelled. Existing systems typically reweight phases without treating phase dependencies as a primary object of learning and interpretation.

The literature also exposes a practical concern: the strongest truly multi-phase liver lesion datasets used in recent work are often private, whereas the public LiTS benchmark is not a dedicated four-phase dataset (Bilic et al., 2023; Liu et al., 2024). This produces a reproducibility gap between the algorithms that are conceptually closest to our problem and the data that are most accessible for public benchmarking.

The methods reviewed span a broad range of strategies. Some emphasise feature aggregation through attention or adaptive fusion; others rely on synthesis to complete missing phases; still others promote cross-phase interaction at the spatial level (Xu et al., 2021; Liu et al., 2024; Wu et al., 2023). Work on missing-view and incomplete-modality learning further shows that robustness can be improved through heterogeneity-aware adaptation, multimodal transformers, or synthesis-based completion (Raju et al., 2020; Zhang et al., 2022; Chen et al., 2024). A consistent pattern emerges: multi-phase evidence is informative, but the mech-

anism by which a model should represent phase agreement, contradiction, or complementarity remains under-specified (Zhang et al., 2021b).

This distinction is central to the present contribution, which is not intended as another attention U-Net variant. The novelty target identified in the survey is to preserve the identity of each phase long enough for the network to reason about cross-phase interactions explicitly, and the proposed model is evaluated against this standard.

This observation directly motivates the proposed design: a model that preserves phase-specific features, reasons over how phases reinforce one another, and exposes interpretable evidence regarding phase importance and phase-subset behaviour.

3. Methodology

The methodology is organised as a staged progression of improvements. Each phase reuses the working components of the previous pipeline and then introduces a more realistic dataset setting, a stronger representation, or a more explicit modelling assumption. The progression follows a pyramidal structure: Phase 3 establishes the baseline segmentation machinery, Phase 4 transforms it into a genuine multi-phase pipeline, and Phase 5 introduces the dependency-aware architecture together with the main dependency experiments.

Phase 3 baseline on LiTS. The corrected baseline implements a conventional 2D U-Net with four encoder stages, a bottleneck, and a symmetric decoder. The input channel count is configurable, but the baseline operates on a single HU-windowed CT slice per example. The LiTS pipeline uses a volume-level split with 35 training volumes, 8 validation volumes, and 8 test volumes, yielding 4,978 training slices, 833 validation slices, and 930 test slices after liver-slice filtering. Training employs a combined cross-entropy and Dice loss, and the corrected test result reaches 0.9111 liver DSC and 0.3360 tumor DSC.

Phase 4 multi-phase Harvard baseline. The next stage retains the segmentation training structure of Phase 3 but changes the problem setting entirely. Instead of single-phase LiTS slices, the pipeline now ingests four distinct CT phases per patient. Each cached sample stores a 4-channel stack corresponding to NC, AP, PVP, and DP, together with a three-class segmentation mask, a phase-presence vector, and patient metadata. This stage introduces TAR extraction, phase identification, cache manifests, patient-wise splitting, and tumor-aware oversampling. The model used in this stage is a four-phase *BaselineAttentionUNet* (Oktay et al., 2018).

Phase 5 dependency-aware model. Phase 5 retains the Harvard multi-phase framework of Phase 4 but upgrades both the data representation and the modelling strategy. The dataset is expanded from 50 to 100 patient archives, and the input representation moves from one slice per phase to a 2.5D formulation with three neighbouring slices per phase, yielding 12 input channels in total. This preserves local axial context without moving to a full 3D pipeline and renders phase reasoning more spatially informed.

The Harvard pipeline also incorporates several practical components absent from the single-phase baseline: automatic data discovery from cloud storage, local archive staging for stable access, cache-metadata validation to prevent stale preprocessed slices, and storage of every training example as a compressed NPZ file containing the image tensor, mask, phase-presence vector, patient identifier, and slice index. These engineering steps form an integral part of the methodology because the final model depends on stable multi-phase alignment and reproducible preprocessing.

Slice construction is a key methodological choice. Rather than sampling all slices uniformly, the pipeline retains tumor-bearing slices wherever possible and supplements them with neighbouring liver-context slices. This strategy improves the effective lesion-to-background ratio during training while preserving sufficient organ anatomy for the decoder to learn coherent liver boundaries.

Architecturally, the transition from Phase 4 to Phase 5 constitutes the main theoretical contribution. The Phase 4 *BaselineAttentionUNet* treats the four phases as a single jointly stacked input stream; its decoder uses attention-gated skip connections, but the feature extractor itself does not preserve phase identity beyond the raw input channels. The proposed *DependencyAttentionUNet* addresses this by introducing a shared per-phase encoder: each phase-specific input group is processed independently to produce a deep feature map and a set of skip tensors prior to fusion.

The shared per-phase encoder is motivated by two considerations. First, it preserves a distinct representation for each phase, which is necessary if phase relationships are to be modelled explicitly rather than absorbed into a generic feature tensor. Second, it keeps the architecture parameter-efficient because the same encoder weights are reused across NC, AP, PVP, and DP. This provides a principled means of comparing phases at the feature level while still learning a common representation space (Zhang et al., 2022).

After encoding, the model computes a scalar gate for each phase from the corresponding bottleneck feature. These gates act as learned phase-importance weights and are used to compute a normalised weighted average of both the bottleneck features and the skip tensors. The implementation

clamps the effective gate range to $[0.15, 1.0]$ to prevent any phase from collapsing entirely during training. As a result, the decoder does not receive a blind concatenation of phase features but a fused representation whose composition reflects what the model has learned about per-sample phase usefulness.

The principal novelty is an explicit pairwise interaction term. For every pair of phases i and j , the model computes an elementwise interaction feature $f_i \odot f_j$. These pairwise interaction maps are averaged, projected back into the fused bottleneck space, and added to the gated fused representation prior to decoding. Conceptually, this encourages the model to learn where phases agree, reinforce one another, or jointly highlight structures that would be weaker in a single stream. This step moves the framework from implicit multi-phase fusion to explicit interaction modelling (Zhang et al., 2021b; Xu et al., 2021; Liu et al., 2024).

The decoder then operates on this interaction-aware fused representation using attention-gated skip information (Oktay et al., 2018). This retains a key strength of the Phase 4 baseline—stronger skip refinement—while altering the information delivered to the decoder. The proposed architecture is therefore not a wholesale replacement of earlier design ideas, but an extension that preserves the useful components and inserts explicit dependency modelling where the prior baseline was weakest.

Loss and inference. The training objective is a tumor-focused combination of focal cross-entropy (Lin et al., 2017), weighted Dice loss, and tumor focal Tversky loss (Abraham & Khan, 2019), mixed at weights 0.25/0.35/0.40. The checkpoint selection criterion is tumor-weighted ($0.35 \cdot \text{liver} + 0.65 \cdot \text{tumor}$). Inference uses horizontal-flip test-time augmentation (TTA), validation-tuned liver and tumor thresholds, and post-processing to remove very small tumor components and fill organ holes. Light phase dropout during training regularises the phase-fusion process and renders the learned representation more amenable to the all-15-subset analysis subsequently performed on the fixed trained model.

3.1. Phase 3: Baseline U-Net on LiTS

Phase 3 establishes a minimum viable segmentation pipeline. The corrected baseline is a standard U-Net operating on one CT slice at a time after application of a liver-focused HU window. Its role is to validate the choice of labels, metrics, training loop, checkpointing logic, and qualitative inspection routines before moving to the more challenging multi-phase setting.

Unlike a slice-level split, the Phase 3 baseline employs a volume-level split, which is substantially safer for medical-image evaluation. The corrected pipeline uses 35 training

volumes, 8 validation volumes, and 8 test volumes, yielding 4,978 training slices, 833 validation slices, and 930 test slices after filtering for liver-containing images. The resulting baseline therefore provides a more reliable reference for project progression, even though it remains on LiTS rather than the Harvard dataset.

Architecturally, the Phase 3 U-Net is a plain encoder–bottleneck–decoder pathway without phase reasoning. The corrected test result is 0.9111 liver DSC and 0.3360 tumor DSC. The role of this stage is therefore conceptual and procedural: it characterises what a straightforward segmentation model can achieve when no explicit phase decomposition or dependency mechanism is available (Li et al., 2018; Isensee et al., 2021).

3.2. Phase 4: Harvard Multi-Phase Baseline

Phase 4 represents the first true alignment between implementation and research objective. The pipeline moves from a public single-phase proxy dataset to Harvard archives containing all four CT phases and phase-specific liver and tumor masks. This stage requires solving not only model-design issues but also data-engineering challenges that are absent from the LiTS baseline.

The Phase 4 preprocessing must identify CT phases from filenames, separate liver and tumor masks, validate volume geometry, and build a cache that can be safely reused across runs. Each resulting NPZ sample stores a 4-channel multi-phase image, a merged 3-class mask, a phase-presence vector, and patient metadata. This is the first stage in which the framework becomes genuinely phase-aware at the data level.

The model employed in Phase 4 is a *BaselineAttentionUNet*. It is already stronger than the plain LiTS U-Net in terms of skip-connection refinement, owing to the use of attention gates in the decoder (Oktay et al., 2018). Nevertheless, it still treats the four phases as a jointly stacked input stream; in other words, Phase 4 is a multi-phase baseline, but not yet a dependency-aware one (Xu et al., 2021; Liu et al., 2024).

Phase 4 also establishes patient-wise train, validation, and test splits, which is methodologically important in medical imaging: when the same patient can appear in multiple slices, patient-level separation is substantially safer than random slice-level splitting. The Harvard baseline therefore represents an improvement in both data realism and evaluation rigour.

3.3. Phase 5: Final Dependency-Aware Submission

Phase 5 is the stage most closely aligned with the survey-defined novelty. The principal conceptual change is that phases are no longer fused immediately; instead, the network preserves phase-specific representations until each

Table 1. Harvard data scale and split progression from the baseline to the final submission.

Property	Phase 4	Phase 5
Patient archives	50	100
Train / val / test pts	34 / 8 / 8	70 / 15 / 15
Train / val / test slices	721 / 153 / 168	2276 / 462 / 474
Input channels	4	12
Phase context	1 slice / phase	2.5D, 3 / phase

phase has been encoded, gated, and incorporated into an explicit interaction term.

The 2.5D input formulation plays a key role: each phase contributes three neighbouring slices, allowing the model to capture local axial continuity within each phase before reasoning across phases. This design provides more context than a single 2D slice while avoiding the computational burden of full 3D segmentation.

Phase 5 also refines the optimisation procedure. Tumor slices are oversampled more aggressively, the loss is shifted toward tumor-sensitive terms, early stopping is introduced, and final test predictions are decoded using validation-tuned thresholds rather than plain argmax. Concretely, the optimisation employs a 0.25/0.35/0.40 mixture of focal cross-entropy, weighted Dice, and tumor focal Tversky losses, together with a tumor oversampling weight of 12.0. These choices reflect the empirical observation that tumor segmentation is substantially harder than liver segmentation and requires more targeted optimisation.

Phase 5 is also where the framework becomes interpretable in a way that directly supports the survey claim. The network reports average phase-gate activations, and a complete combinational dependency analysis is performed on the fixed trained model by evaluating all 15 possible non-empty phase combinations. The framework therefore provides not only final Dice values but direct evidence regarding phase dependence, including higher-order three-phase behaviour.

4. Results

4.1. Phase-by-Phase Result Progression

The clearest empirical trend is the transition from the four-phase Harvard baseline to the proposed dependency-aware system. The Harvard baseline attains 0.8896 liver DSC and 0.5287 tumor DSC on the 50-case setup, while the proposed system attains 0.9435 liver DSC and 0.6975 tumor DSC on the 100-case setup. Because Phase 5 also introduces 2.5D context, stronger slice selection, richer losses, tuned thresholds, and post-processing, this gain is best interpreted as a pipeline-level improvement rather than the effect of a single isolated module.

Table 2. Final experimental setup and inference choices (Phase 5).

Setting	Value
Phases	NC, AP, PVP, DP
Input format	2.5D, 3 slices per phase
Total channels	12
Batch size	8
Epoch budget	24
Optimizer	AdamW
LR / weight decay	10^{-4} / 10^{-3}
Loss weights	0.25 / 0.35 / 0.40
Tumor oversampling	12.0
Gate clamp	[0.15, 1.0]
Checkpoint score	$0.35 \cdot \text{liver} + 0.65 \cdot \text{tumor}$
Tumor threshold	0.24
Liver threshold	0.50
Min tumor pixels	32

Table 3. Architecture comparison between the Harvard baseline and the final model.

Property	Phase 4 baseline	Phase 5 final
Input representation	4 ch, 1 slice/phase	12 ch, 3 slices/phase
Encoder strategy	Single-stream	Shared per-phase
Phase weighting	Implicit	Explicit learned gates
Interaction modeling	None	Pairwise cross-phase
Missing-phase handling	Presence vector	Presence + phase dropout
Inference tuning	Direct evaluation	Threshold + TTA + post-proc.

Figure 1 shows the training dynamics of the proposed 2.5D dependency model: validation tumor DSC stabilises before early stopping triggers, while liver DSC plateaus in the high 0.94 range. Figure 2 summarises the held-out test DSC after tuned inference.

Even with this caveat, the direction of change is meaningful. The corrected Phase 3 baseline characterises what a plain single-phase U-Net can achieve on LiTS under a safer volume-level split, while Phase 4 illustrates the difficulty of moving to a realistic four-phase private dataset. Phase 5 then improves substantially over the Harvard baseline while operating in the most clinically structured setting of the three. This indicates that the gains arise not only from improved optimisation but also from a more appropriate representation of multi-phase evidence.

Qualitatively, each stage addresses a distinct problem. Phase 3 addresses the basic segmentation problem. Phase 4 addresses the dataset and multi-phase ingestion problem. Phase 5 addresses the explicit dependency-modelling problem while also strengthening the surrounding optimisation and inference pipeline.

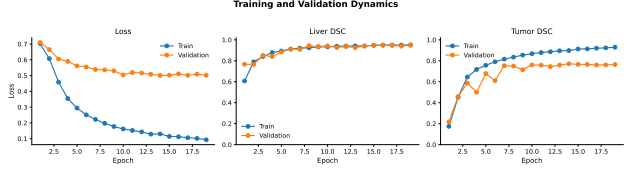


Figure 1. Training dynamics for the final 2.5D dependency model. Train/validation loss and validation liver and tumor DSC over epochs. Validation tumor DSC stabilises before early stopping triggers, while liver DSC plateaus in the high 0.94 range. Early stopping uses patience 5 on the combined validation score $0.35 \cdot \text{liverDSC} + 0.65 \cdot \text{tumorDSC}$.

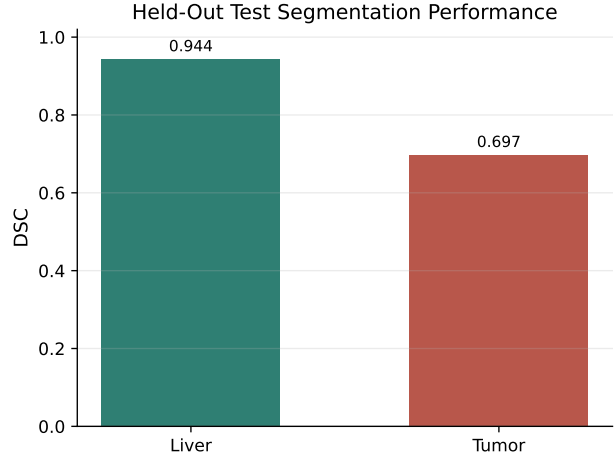


Figure 2. Held-out test DSC of the proposed 2.5D dependency model on the 100-patient slice-level test split. Liver DSC reaches 0.9435 and tumor DSC 0.6975.

The proposed framework also exposes interpretable phase-importance values through the average learned gate activations on the test set. The mean gate strengths are 0.6709 for NC, 0.5903 for AP, 0.6134 for PVP, and 0.6291 for DP (Figure 3). These values indicate that the network does not weight all phases uniformly and that the multi-phase representation remains phase-aware throughout the forward pass.

From an experimental standpoint, this is informative because the model never receives phase labels as a direct segmentation target; the gate values emerge from training as auxiliary evidence of what the network finds useful. In other words, although the model is trained solely to segment, it learns a measurable phase-usage profile as a by-product of its architecture.

To assess whether the model has learned meaningful dependency behaviour rather than simply benefiting from additional channels, we conduct a complete combinational dependency analysis on the fixed trained model. All 15 possi-

Table 4. Project progression across the three implementation stages.

Stage	Setting	Liver	Tumor
Phase 3	LiTS, 1-channel 2D U-Net	0.9111	0.3360
Phase 4	Harvard-50, 4-phase Att. U-Net	0.8896	0.5287
Phase 5	Harvard-100, 2.5D Dependency	0.9435	0.6975

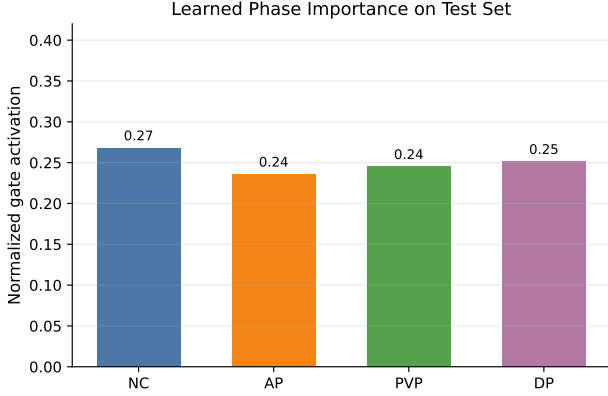


Figure 3. Average phase-gate activations on the final Harvard-100 test set. Gates are clamped to $[0.15, 1.0]$ to prevent phase deletion during training. NC is slightly emphasized and AP slightly down-weighted, but all four phases receive non-trivial weight.

ble non-empty subsets of the four phases are evaluated on the test set, covering single-phase, two-phase, three-phase, and all-phase configurations. The strongest single phase is PVP with tumor DSC 0.3353. The strongest two-phase combination is PVP+DP with tumor DSC 0.7209, and the strongest three-phase combination is AP+PVP+DP with tumor DSC 0.7109. Both exceed the all-phase configuration, which attains tumor DSC 0.6975 (Table 5, Figures 4–5).

4.2. Interpretation of the Quantitative Results

The liver Dice of 0.9435 indicates that the organ segmentation branch is strong and stable. This is non-trivial because the inference pipeline explicitly constrains tumor decoding to plausible liver regions; more accurate liver masks therefore support tumor localisation indirectly.

The tumor Dice of 0.6975 should be interpreted in the context of task difficulty. Tumors are smaller, morphologically diverse, and substantially more imbalanced than liver tissue. Furthermore, the Harvard experiments involve all four phases, which poses a stronger representation problem than the single-phase LiTS baseline. Within that context, the Phase 5 tumor result represents a meaningful improvement over the Harvard baseline and is considerably stronger than the corrected single-phase baseline.

Table 5. Complete combinational dependency analysis on the fixed trained model across all 15 possible non-empty phase combinations.

Input phases	N	Liver	Tumor
NC	1	0.9183	0.1857
AP	1	0.9320	0.3113
PVP	1	0.9530	0.3353
DP	1	0.9372	0.1935
NC+AP	2	0.9356	0.5696
NC+PVP	2	0.9367	0.7137
NC+DP	2	0.9212	0.5655
AP+PVP	2	0.9456	0.6863
AP+DP	2	0.9399	0.6635
PVP+DP	2	0.9322	0.7209
NC+AP+PVP	3	0.9459	0.6887
NC+AP+DP	3	0.9400	0.6394
NC+PVP+DP	3	0.9344	0.6985
AP+PVP+DP	3	0.9440	0.7109
NC+AP+PVP+DP	4	0.9435	0.6975

The complete phase-combination table adds a further layer of interpretation. It shows that the model does not require every phase equally and that combinations containing PVP dominate the ranking. This finding is consistent with the clinical importance of portal venous contrast in lesion characterisation and with the survey position that phase relationships should be studied directly rather than concealed within an undifferentiated input tensor.

The analysis also covers higher-order combinations rather than singles and pairs alone. The best three-phase subset, AP+PVP+DP, attains tumor DSC 0.7109, and a further three-phase subset, NC+PVP+DP, attains 0.6985; both indicate that higher-order dependency patterns can be competitive with, or superior to, the full four-phase configuration.

4.3. What the Phase Analysis Tells Us

The subset experiments are particularly valuable because they elevate the analysis beyond a performance report. Evaluating the trained network under restricted phase combinations reveals which parts of the learned representation are genuinely useful and which are merely available. Because the analysis covers all 15 non-empty subsets of the four phases, it constitutes a full mapping of model behaviour over the available phase set rather than a limited ablation.

Equally important, the full four-phase result is not the best tumor score. This observation does not weaken the contribution; rather, it reinforces the argument that dependency-aware segmentation should analyse combinations explicitly. The best two-phase subset, PVP+DP, attains 0.7209 tumor DSC, and the best three-phase subset, AP+PVP+DP, attains 0.7109, both surpassing the all-phase score of 0.6975. Were every added phase to improve results monotonically, there would be little reason to study dependencies. The fact that

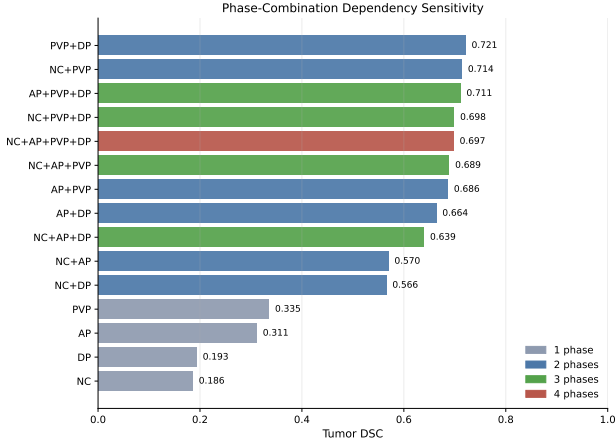


Figure 4. Tumor Dice ranking induced by the complete combinational dependency analysis on the fixed trained model. PVP-centered subsets dominate, and both the best two-phase and best three-phase subsets outperform the all-phase setting.

several reduced subsets outperform the full stack indicates that phase interaction is selective rather than uniform.

This observation also clarifies why the proposed architecture is methodologically warranted. A model that merely concatenates phases can benefit from additional information, but it cannot communicate whether that information is synergistic, redundant, or distracting. The proposed framework begins to address this question by combining learned-gate analysis with a complete higher-order subset evaluation.

5. Discussion

5.1. Why the Final Method Works

The proposed system supports the central thesis of the prior survey: it is beneficial to preserve phase-specific information long enough to model interactions explicitly. The observed gains are not explained by a generic “more input channels” account; the model learns differentiated importance for NC, AP, PVP, and DP, and responds predictably to phase subset removal (Xu et al., 2021; Zhang et al., 2022).

The dependency study is particularly informative because it is a complete combinational dependency analysis on the fixed trained model. Rather than testing only a small set of intuitive ablations, the analysis evaluates all 15 non-empty phase combinations. This substantially strengthens the findings: the framework demonstrates not only that phase information matters, but also how performance varies across single-phase, pairwise, and higher-order combinations.

The observation that PVP+DP and AP+PVP+DP both out-

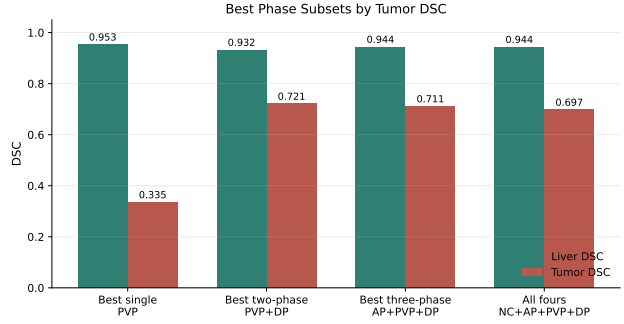


Figure 5. Best phase subset per cardinality. PVP is the strongest single phase (tumor DSC 0.3353); PVP+DP is the strongest two-phase subset (0.7209); AP+PVP+DP is the strongest three-phase subset (0.7109); the four-phase input remains competitive (0.6975) but is not the absolute top tumor subset in this run.

perform the full four-phase configuration indicates that dependency-aware segmentation is also a selection problem. Some phases are more informative for tumor delineation than others, and the strongest behaviour arises from learning how phase evidence should be combined rather than forcing the model to treat all views as equally useful.

The results are clinically plausible. PVP and DP preserve lesion-to-parenchyma contrast and delayed-washout behaviour that are central to liver lesion interpretation under LI-RADS-style reasoning (Chernyak et al., 2018). This grounds the dependency scores and subset trends in a clinically meaningful manner that an opaque feature-fusion mechanism cannot, and it is consistent with recent multi-phase liver lesion analysis literature (Wu et al., 2024; 2023).

A further observation from the staged progression is that preprocessing strategy matters almost as much as architecture. The Harvard phases required archive parsing, phase matching, cache validation, patient-level split protection, and selective tumor-focused slice construction before the model could make effective use of the data. Without this infrastructure, the dependency-aware model would not have had a stable multi-phase learning problem to address.

The framework also illustrates that evaluation should not be reduced to a single Dice value. The tuned inference thresholds, gate activations, and phase-subset experiments all contribute to explaining why the proposed method works, supporting the survey position that robustness and interpretability should be treated as first-class evaluation targets in multi-phase liver segmentation (Hu et al., 2024).

Figure 6 presents representative qualitative segmentation examples from the held-out test split. These are illustrative high-agreement cases drawn from the trained model on real held-out test samples; quantitative DSC remains the pri-

mary evidence.

A useful supplementary observation is that longer or alternate-seed runs remain consistent with the headline results. A longer single-seed run attained 0.9529 liver DSC, 0.7371 tumor DSC, with validation tumor DSC 0.7835. A two-seed sanity check reported seed 123 at 0.9198 liver and 0.6381 tumor, with a mean $\pm$ std of 0.9317 ± 0.0168 for liver and 0.6678 ± 0.0420 for tumor. Although these are not the main headline numbers, they provide additional evidence that the proposed design retains headroom and does not depend on a single unusually favourable run.

5.2. Limitations and Remaining Gaps

Several limitations of this work should be acknowledged. The LiTS baseline and the Harvard experiments are not directly comparable, and their numerical difference should not be interpreted as a controlled benchmark. Their role is to document methodological progression rather than to constitute a one-to-one leaderboard comparison.

The Harvard slice cache is enriched in tumor-bearing slices through tumor-positive oversampling but still contains a substantial fraction of tumor-negative liver-context slices. In the final train split, the pipeline reports 938 tumor-positive and 1338 tumor-negative slices. Generalisation to volumes with very few or no lesions is therefore not evaluated directly.

The principal evaluation is slice-level rather than full 3D volume-level. The paper does not establish whether the same gains hold under volumetric consistency metrics or whole-volume post-processing.

The Harvard data available to this study do not expose tumor-type labels such as HCC, ICC, BCLM, CRLM, or hemangioma; consequently, no tumor-subtype performance claims are made. There is also no external validation cohort beyond the internal Harvard split.

The headline result is obtained from a single patient-wise train / validation / test split. The two-seed sanity check is encouraging but is not a substitute for a full repeated-split study.

Missing phases were simulated through phase dropout and inference-time subset masking rather than collected as naturally incomplete acquisitions. Furthermore, the all-15-subset analysis is performed on the fixed trained model and should not be conflated with 15 separately retrained baselines (Raju et al., 2020; Zhang et al., 2022; Chen et al., 2024).

The Dice score itself requires ground-truth annotations; consequently, while the results are clinically suggestive, the current pipeline is evaluated in a supervised benchmarking setting rather than a deployment setting without reference

masks.

The explicit interaction term is parameterised pairwise: the model multiplies pairs of per-phase feature maps and projects them back into the fused bottleneck. Three- and four-way coupling is therefore captured only implicitly through the decoder, not as an explicit higher-order tensor term. Higher-order behaviour is, however, exposed at inference time by the all-15-subset sensitivity analysis.

5.3. Alignment with the Original Project Aim

The contribution of this work is best assessed against the prior survey that defined the research direction. By that standard, the proposed framework is well aligned with the intended novelty. The architecture is multi-phase, phase-aware, dependency-aware, and partially interpretable. It explicitly models pairwise phase interactions rather than relying on implicit fusion within a shared feature extractor, and it validates these interactions through a complete combinatorial dependency analysis on the fixed trained model.

At the same time, this work remains deliberately measured in its claims. The present implementation demonstrates explicit interaction modelling but does not exhaust the strongest ambitions of the survey concerning structured higher-order dependencies and clinically realistic missing-phase robustness. This nuance is a strength rather than a weakness, since it preserves a clear distinction between what has been implemented, what has been validated, and what remains for future work.

6. Conclusion

This work has progressed from a standard U-Net baseline to a full dependency-aware multi-phase CT segmentation pipeline. The proposed model aligns well with the research direction set by the prior survey, preserving phase-specific representations, learning phase-importance gates, and explicitly modelling pairwise cross-phase interactions prior to decoding.

On the 100-patient Harvard experiment, the model attains 0.9435 liver DSC and 0.6975 tumor DSC. The principal finding, however, is the complete combinatorial dependency analysis on the fixed trained model. Across all 15 possible non-empty phase combinations, the best two-phase subset, PVP+DP, attains tumor DSC 0.7209 and the best three-phase subset, AP+PVP+DP, attains 0.7109, both surpassing the all-phase configuration.

The most important directions for future work are a strictly controlled baseline-versus-dependency comparison under identical final settings, evaluation on naturally incomplete phase patterns, and richer interaction modules that move beyond pairwise feature products. Such extensions would

Representative Multiphase Inputs and Tumor Segmentation Outputs

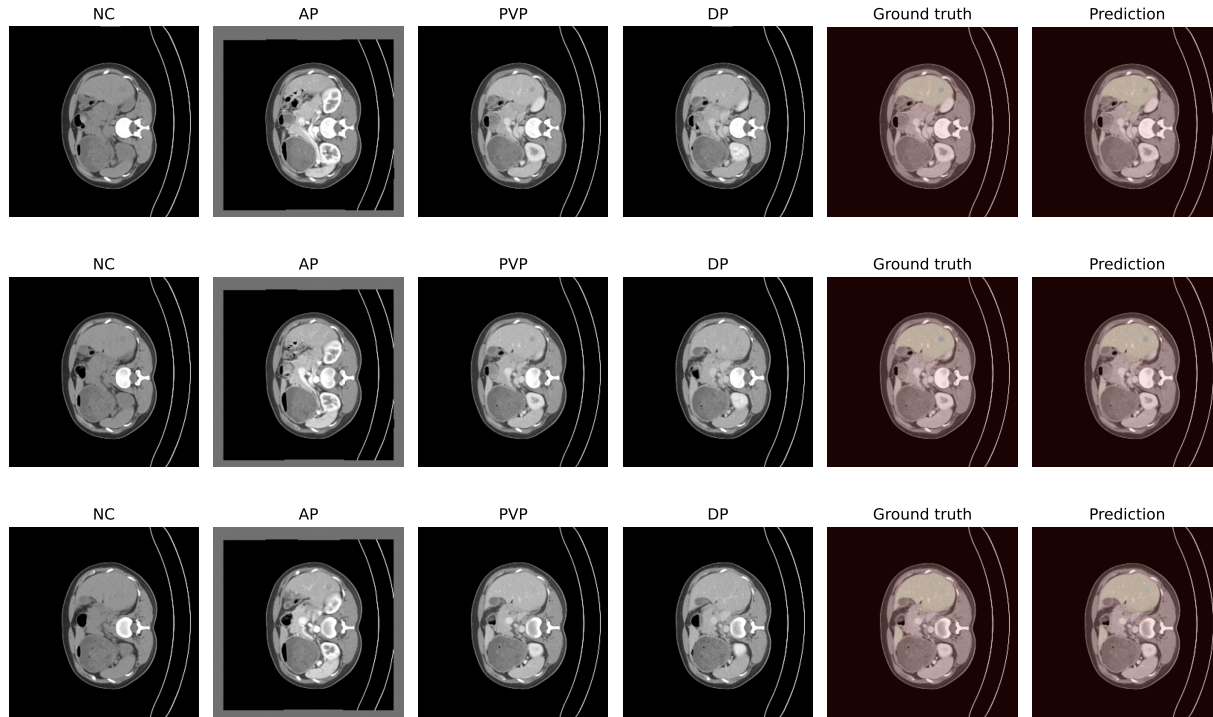


Figure 6. Representative qualitative segmentation examples from the held-out test split, produced by the proposed 2.5D dependency model after TTA, threshold tuning, and liver-constrained post-processing. Each row shows multi-phase input slices, the ground-truth multiclass mask (background / liver / tumor), and the predicted mask. The examples are illustrative high-agreement cases on real held-out test samples; quantitative DSC remains the primary evidence (Table 5).

bring the framework closer to the full research vision proposed in the original survey.

Even in its current form, the work demonstrates a coherent research progression: a baseline segmentation system, a realistic multi-phase adaptation, and an explicit interaction model whose behaviour is supported by quantitative segmentation results, higher-order subset findings, and interpretable dependency analysis.

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